

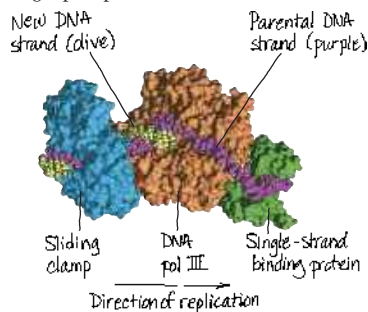
Summary of Key Concepts Questions

16.1 Each strand in the double helix has polarity; the end with a phosphate group on the 5' carbon of the sugar is called the 5' end, and the end with an —OH group on the 3' carbon of the sugar is called the 3' end. The two strands run in opposite directions, one running 5' → 3' and the other alongside it running 3' → 5'. Thus, each end of the molecule has both a 5' and a 3' end, one on each strand of the double helix. This arrangement is called antiparallel. If the strands were parallel, they would both run 5' → 3' in the same direction, so an end of the molecule would have either two 5' ends or two 3' ends. **16.2** On both the leading and lagging strands, DNA polymerase adds onto the 3' end of an RNA primer synthesized by primase, synthesizing DNA in the 5' → 3' direction. Because the parental strands are antiparallel, however, only on the leading strand does synthesis proceed continuously into the replication fork. The lagging strand is synthesized bit by bit in the direction away from the fork as a series of shorter Okazaki fragments, which are later joined together by DNA ligase. Each fragment is initiated by synthesis of an RNA primer by primase as soon as a given stretch of single-stranded template strand is opened up. Although both strands are synthesized at the same rate, synthesis of the lagging strand is delayed because initiation of each fragment begins only when sufficient template strand is available. **16.3** The chromatin in an interphase nucleus is present as the 10-nm fiber, either fairly loosely arranged in euchromatin or more densely arranged in heterochromatin (such as at the centromeres and telomeres). The euchromatin is also subdivided into larger compartments and smaller looped domains. This organization may reflect differences in gene expression occurring in these regions.

Test Your Understanding

1. C 2. C 3. B 4. D 5. A 6. D 7. B 8. A 9. Like histones, the *E. coli* proteins would be expected to contain many basic (positively charged) amino acids, such as lysine and arginine, which can form weak bonds with the negatively charged phosphate groups on the sugar-phosphate backbone of the DNA molecule.

11.



Chapter 17

Figure Questions

Figure 17.3 The previously presumed pathway would have been wrong. The new results would support this pathway: precursor → citrulline → ornithine → arginine. They would also indicate that class I mutants have a defect in the second step and class II mutants have a defect in the first step. **Figure 17.5** The mRNA sequence (5'-UGGUUUGGCUCA-3') is the same as the nontemplate DNA strand sequence (5'-TGGTTTGGCTCA-3'), except there is a U in the mRNA wherever there is a T in the DNA. The nontemplate strand is probably used to represent a DNA sequence because it so closely resembles the mRNA sequence, containing codons. (This is why it's called the coding strand.) **Figure 17.6** Arg (or R)—Glu (or E)—Pro (or P)—Arg (or R) **Figure 17.8** The processes are similar in that polymerases form polynucleotides complementary to an antiparallel DNA template strand. In replication, however, both strands act as templates, whereas in transcription, only one DNA strand acts as a template. This reflects, of course, the fact that replication results in a double-stranded product (DNA) while transcription results in a single-stranded product (an RNA). **Figure 17.9** The RNA polymerase would bind directly to the promoter, rather than being dependent on the previous binding of transcription factors.

Figure 17.12

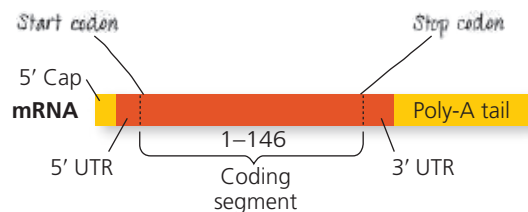


Figure 17.16 The anticodon on the tRNA is 3'-AAG-5', so it would bind to the mRNA codon 5'-UUC-3'. This codon codes for phenylalanine (Phe, or F), which is the amino acid this tRNA would carry. **Figure 17.22** It would be packaged in a vesicle, transported to the Golgi apparatus for further processing, and then transported via a vesicle to the plasma membrane. The vesicle would fuse with the membrane, releasing the protein outside the cell. **Figure 17.24** The mRNA farthest to the right (the longest one) started transcription first. The ribosome at the top, closest to the DNA, started translating first and thus has the longest polypeptide.

Concept Check 17.1

- Recessive
- A polypeptide made up of 10 Gly (glycine) amino acids
-

Nontemplate sequence (from template sequence given in problem), written 3' → 5': 3'-ACCACTGAA-5'

If above used as template, mRNA sequence: 5'-UGUGAUAUU-3'

Translated: Cys-STOP

If the nontemplate sequence could have been used as a template for transcribing the mRNA, the protein translated from the mRNA would have a completely different amino acid sequence, so it would not be able to function as the original protein (translated from an mRNA transcribed from the template strand). It would also be shorter because of the UGA stop signal shown in the mRNA sequence above—and possibly others earlier in the mRNA sequence.

Concept Check 17.2

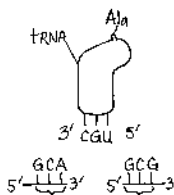
- A promoter is the region of DNA to which RNA polymerase binds to begin transcription. It is at the upstream end of the gene (transcription unit).
- In a bacterial cell, part of the RNA polymerase recognizes the gene's promoter and binds to it. In a eukaryotic cell, transcription factors must bind to the promoter first, then the RNA polymerase binds to them. In both cases, sequences in the promoter determine the precise binding of RNA polymerase so the enzyme is in the right location and orientation.
- The transcription factor that recognizes the TATA sequence would be unable to bind, so RNA polymerase could not bind and transcription of that gene most likely would not occur.

Concept Check 17.3

- Due to alternative splicing of exons, each gene can result in multiple different mRNAs and can thus direct synthesis of multiple different proteins.
- In watching a pre-recorded show, you watch segments of the show itself (exons) and fast-forward through the commercials, which are thus like introns. However, unlike introns, commercials remain in the recording, while the introns are cut out of the RNA transcript during RNA processing.
- Once the mRNA has exited the nucleus, the cap prevents it from being degraded by hydrolytic enzymes and facilitates its attachment to ribosomes. If the cap were removed from all mRNAs, the cell would no longer be able to synthesize any proteins and would probably die.

Concept Check 17.4

- First, each aminoacyl-tRNA synthetase specifically recognizes a single amino acid and attaches it only to an appropriate tRNA. Second, a tRNA charged with its specific amino acid binds only to an mRNA codon for that amino acid.
- A signal peptide on the leading end (amino end, or N-terminus) of the polypeptide being synthesized is recognized by a signal-recognition particle that brings the ribosome to the ER membrane. There, the ribosome attaches and continues to synthesize the polypeptide, depositing it in the ER lumen.
- Because of wobble, the tRNA could bind to either 5'-GCA-3' or 5'-GCG-3', both of which code for alanine (Ala, or A). Alanine would be attached to the tRNA (see diagram, upper right).
- When one ribosome terminates translation and dissociates, the two subunits would be very close to the cap. This could facilitate their rebinding and initiating synthesis of a new polypeptide, thus increasing the efficiency of translation.



Concept Check 17.5

- In the mRNA, the reading frame downstream from the deletion is shifted, leading to a long string of incorrect amino acids in the polypeptide, and in most cases, a stop codon will occur, leading to premature termination. The polypeptide will most likely be nonfunctional.
- Heterozygous individuals, said to have sickle-cell trait, have a copy each of the wild-type allele and the sickle-cell allele. Both alleles will be expressed, so these individuals will have both normal and sickle-cell hemoglobin molecules. Apparently, having a mix of the two forms of β -globin has no effect under most conditions, but during prolonged periods of low blood oxygen (such as at higher altitudes), these individuals can show some signs of sickle-cell disease.
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Normal DNA sequence (template strand is on top):	3'-TACTTGTCCGATATC-5'
	5'-ATGAACAGGCTATAG-3'
mRNA sequence:	5'-AUGAACAGGUAUAG-3'
Amino acid sequence:	Met-Asn-Arg-Leu-STOP
Mutated DNA sequence (template strand is on top):	3'-TACTTGTCCAATATC-5'
	5'-ATGAACAGGTTATAG-3'
mRNA sequence:	5'-AUGAACAGGUUAUAG-3'
Amino acid sequence:	Met-Asn-Arg-Leu-STOP